

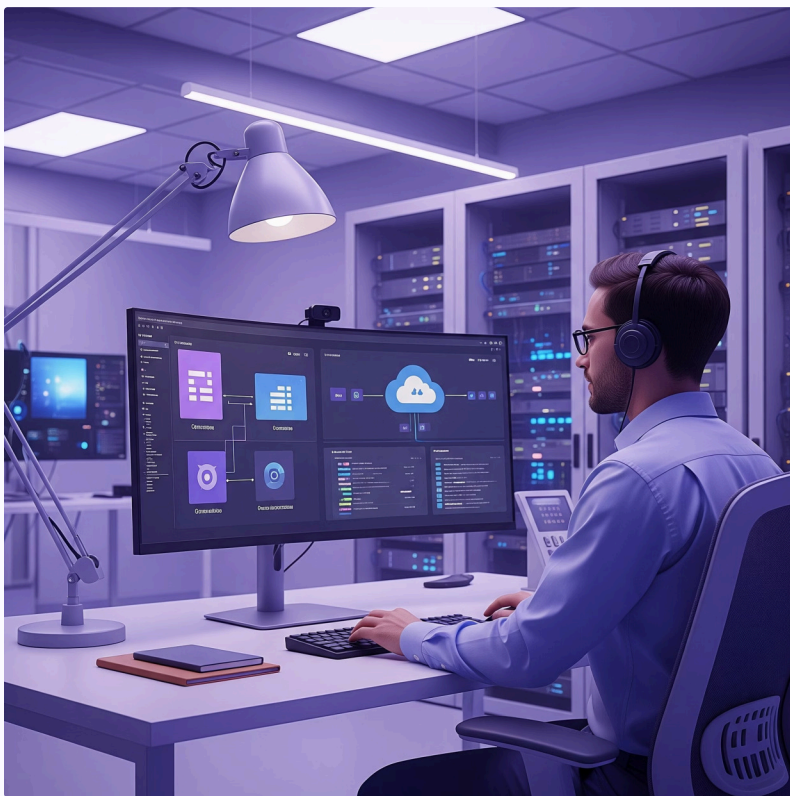
Hybrid On-Prem + Cloud AI Setups

Blending local infrastructure with cloud scalability

A strategic approach for enterprise AI/ML deployments that balances compliance, performance, and cost optimization



Why Hybrid AI?



Enterprises rarely go **all-in cloud** or **all-in on-prem**

Data Residency & Compliance

Healthcare, finance, and government sectors with strict regulatory requirements

Latency-Sensitive Inference

Edge computing, retail applications, and factory floor operations

Cost Control

Leveraging existing GPU/CPU infrastructure investments

Burst Workloads

Using cloud for elastic scaling during peak computation periods

Typical Hybrid Patterns

Strategic workload distribution between on-premises and cloud environments



Training On-Prem, Serving in Cloud

Sensitive data stays local, inference endpoints distributed globally



Inference On-Prem/Edge, Retraining in Cloud

Low-latency predictions locally, model improvement in cloud



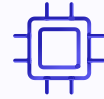
Cloud Bursting

Extra training capacity during peak demands without permanent infrastructure

Split by Data Sensitivity

Process sensitive data on-premises while leveraging cloud for anonymized datasets

On-Prem Components



Compute Infrastructure

GPU/CPU clusters (bare metal or VMware)



Storage

Private object storage (Ceph, MinIO, NetApp)



Orchestration

Kubernetes (OpenShift, Rancher, vanilla k8s)



Security

Enterprise IAM + security gateways

Cloud Components

Elastic GPU Training

Managed services for on-demand ML workloads:

- AWS Sagemaker
- Google Cloud Vertex AI
- Azure Machine Learning

Scalable Model Serving

Dynamic inference endpoints:

- Kubernetes-based deployments
- Managed AI APIs
- Serverless functions

Global Distribution

- Content Delivery Networks
- Global DNS services
- Load balancers

Managed Data Services

- Feature stores
- Vector databases
- Monitoring solutions



Networking & Connectivity

Secure, reliable connections are the foundation of hybrid AI architectures

Private Interconnects

Dedicated, high-bandwidth connections:

- AWS Direct Connect
- Azure ExpressRoute
- Google Cloud Interconnect

VPN Tunnels

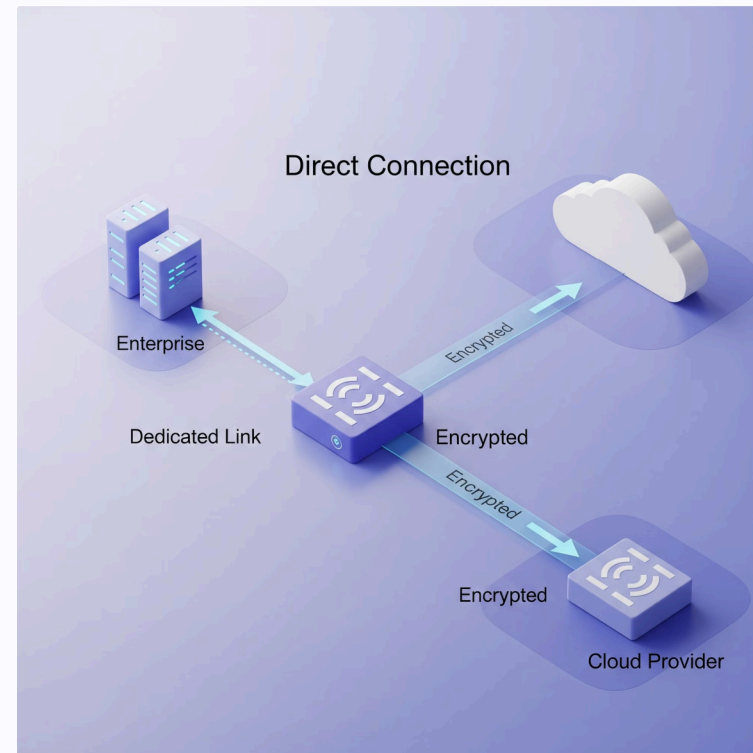
Encrypted connections for smaller deployments or secondary paths

Service-to-Service Auth

Mutual TLS (mTLS) across environment boundaries

Unified Service Discovery

Consistent DNS + routing layer for hybrid endpoints



Data Flow Example

1

1. Data Collection

Raw data stored **on-prem** for compliance requirements

2

2. Feature Engineering

Preprocessed/derived features pushed to **cloud feature store**

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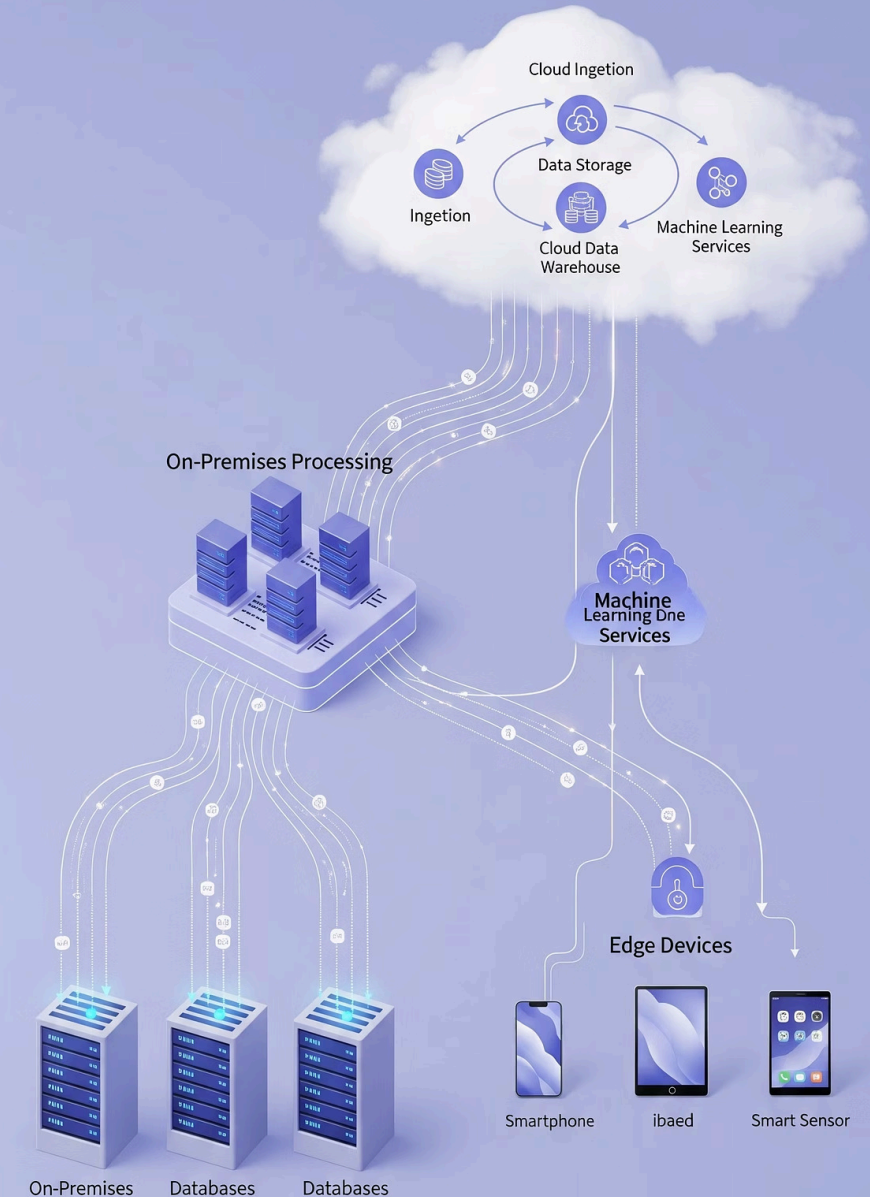
3. Model Training

Models trained in cloud on anonymized/augmented data with elastic resources

4

4. Deployment

Model artifacts synced back to on-prem for local inference



Hybrid AI Challenges

Data Gravity

Moving large datasets across boundaries is expensive and time-consuming

Observability Gaps

Complexity in monitoring and debugging across environments

Security Inconsistency

Disparate IAM systems and secrets management approaches

Performance Impact

Latency introduced by cross-environment network calls

Vendor Lock-in Risk

Dependency on cloud-specific ML services and APIs



⚠ These challenges can significantly impact project timelines and operational costs if not addressed early in the architecture phase.

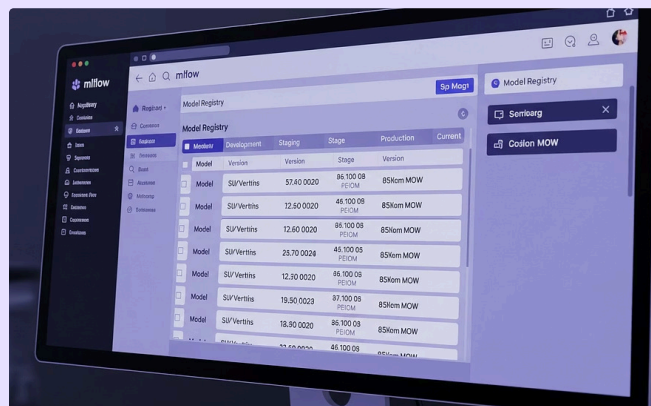
Tools & Frameworks

Key technologies that enable seamless hybrid AI operations



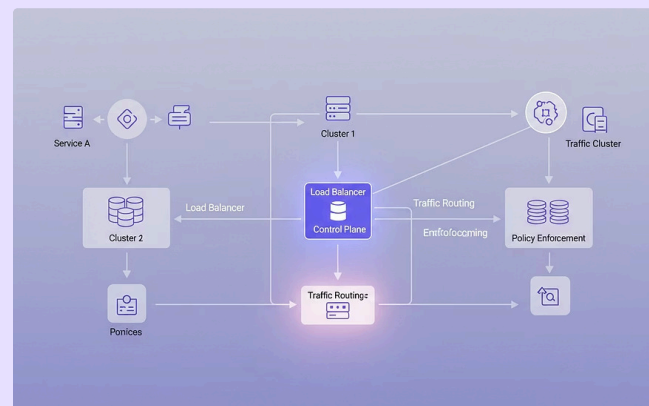
Kubeflow

Portable ML pipelines that run consistently across hybrid infrastructure



MLflow

Hybrid-friendly model registry and experiment tracking



Service Mesh

Istio/Linkerd for traffic management and security across clusters

Data Fabrics

Unified access layer for on-prem + cloud data

Hybrid Kubernetes Management

Google Anthos / Azure Arc for consistent operations

Best Practices

Data Locality

Place workloads **near the data** to reduce egress costs and latency

Consistent Infrastructure

Use **identical containerization & IaC** approaches across environments

Unified Security

Standardize secrets + IAM (HashiCorp Vault, cloud KMS)

End-to-End Observability

Monitor **complete pipeline health** with Prometheus + OpenTelemetry

Resilience Planning

Test **failover paths** between cloud & on-prem regularly



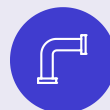
"The key to hybrid success is designing for portability from day one."

Key Takeaways



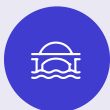
Hybrid = Balance

Compliance + latency + cost + flexibility advantages in one architecture



Cross-Environment Pipelines

AI workflows span **on-prem (data) + cloud (scale)** boundaries



Portability is Critical

Kubernetes & MLOps stacks provide the bridge for workload mobility



Success Factors

Strong networking, unified IAM, portable pipelines, and observability

Questions? Contact our enterprise architecture team at enterprise-arch@company.com